

Series of functions



Outline

Series of functions and Taylor series

The Taylor series of the exponential function

The Taylor series of the elementary trigonometric functions



Series of functions and Taylor series



Introduction: Series of functions and Taylor series

- ▶ In this section we move from numerical series to **series of functions**, that is, series whose terms depend on a variable x
- ▶ We focus on **Taylor series**: they extend to infinity the Taylor polynomial we already know
- ▶ We will understand **when** a function coincides with the sum of its Taylor series: this allows us to compute functions such as e^x , $\sin x$, $\cos x$ to any desired accuracy

Series of functions

- ▶ **Series of functions** are numerical series depending on a parameter x : when these series converge for every value x belonging to a certain interval, they represent **functions of a new kind**
- ▶ There are two important classes of series of functions, namely trigonometric series and power series. We will only deal with **power series**, also called **Taylor series**, which can be seen as a natural extension of Taylor expansions.

Taylor's formula with Lagrange remainder

- Taylor's expansion/formula with Lagrange remainder can be rewritten in the form:

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + E_n(x) \quad (1)$$

where the Lagrange approximation error is

$$E_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} (x - x_0)^{n+1} \quad (2)$$

and c is a suitable point between x_0 and x .

Letting n grow without bound, the sum in Taylor's formula becomes a series: the Taylor series.

Definition 1: Taylor series

If a function f has derivatives of every order, the series

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

*is called the **Taylor series** of the function f centered at x_0 .*

Convergence of the Taylor series

At a given point $x \in (a, b)$, if the error

$$E_n(x) \rightarrow 0 \quad \text{as} \quad n \rightarrow +\infty$$

then the Taylor series is convergent and its sum equals $f(x)$. This is equivalent to saying that the n -th partial sum of the series has a finite limit and this limit is precisely $f(x)$. In formulas:

$$\underbrace{T_{n,x_0}(x)}_{\text{Taylor polynomial}} = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k \rightarrow f(x) \quad \text{as} \quad n \rightarrow +\infty$$

Functions expandable in Taylor series

Definition 2: function expandable in Taylor series on an interval

Given a function $f : D \rightarrow \mathbb{R}$ with derivatives of every order, if

$$E_n(x) \rightarrow 0 \text{ as } n \rightarrow +\infty, \quad \forall x \in I \subseteq D$$

we say that $f(x)$ is **expandable in Taylor series** on the interval I .

Functions expandable in Taylor series

- ▶ There are (infinitely differentiable) functions that are expandable in Taylor series on the whole real line, others that are expandable only on a bounded interval, and others for which the interval reduces to a single point (the point x_0).
- ▶ We now deal with the three elementary transcendental functions:

$$f(x) = e^x, \quad f(x) = \sin x \quad \text{and} \quad f(x) = \cos x$$

which we will show to be expandable in Taylor series on the whole of $\mathbb{R}$. They are three transcendental (not algebraic) functions.

Algebraic and transcendental functions

- ▶ Real functions of a real variable are classified into **algebraic functions and transcendental functions**.
- ▶ Algebraic functions are built through a finite number of applications of the four arithmetic operations, of raising to a power and of extracting the n -th root.
- ▶ Transcendental functions are all the functions that are not algebraic.



Summary

Series of functions:

- ▶ Numerical series depending on a parameter x ; we study **power series** (Taylor series)

Taylor series:

- ▶ $\sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$, for f with derivatives of every order
- ▶ It converges to $f(x)$ if and only if the Lagrange error $E_n(x) \rightarrow 0$ as $n \rightarrow +\infty$
- ▶ f is **expandable** on I if $E_n(x) \rightarrow 0$ for every $x \in I$

The Taylor series of the exponential function



Introduction: The Taylor series of the exponential function

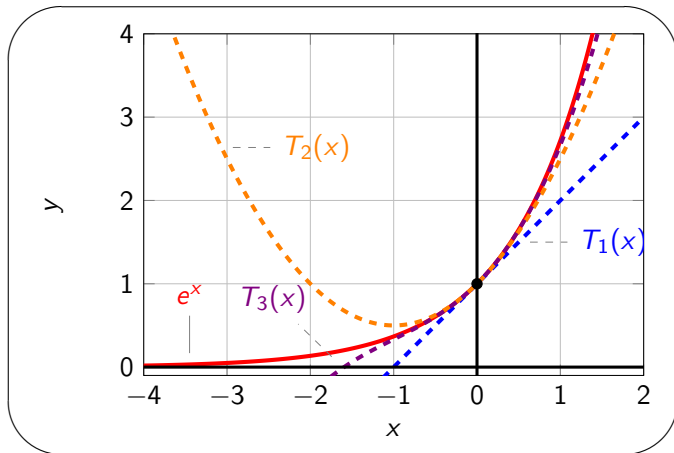
- ▶ In this section we prove that e^x is the sum of its Taylor series **for every** $x \in \mathbb{R}$
- ▶ The key idea is to show that the Lagrange remainder tends to zero, using the hierarchy of infinities
- ▶ As a consequence we obtain a **new formula for the number** e , much more efficient for computing its approximate value

Example (1/2)

Maclaurin formula/expansion of order n of the exponential with Lagrange remainder:

$$\begin{aligned}e^x &= T_n(x) + \frac{e^c}{(n+1)!} x^{n+1} \\&= 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + \frac{e^c}{(n+1)!} x^{n+1} \\&\text{with } c \in [0, x]\end{aligned}$$

Example (2/2)



Taylor series of the exponential function

Observation 1: Taylor series of the exponential function

$$\forall x \in \mathbb{R}, \quad e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

Proof (1/2)

From the Maclaurin expansion of order n of e^x with Lagrange remainder, we have that for every integer n and $x \in \mathbb{R}$ there exists a point c , lying between 0 and x , such that:

$$e^x = \sum_{k=0}^n \frac{x^k}{k!} + \frac{e^c}{(n+1)!} x^{n+1}$$

We fix x and let n tend to $+\infty$. The point c may vary with n but, since it always lies between 0 and x , we have:

$$e^c \leq \begin{cases} e^x & \text{if } x > 0 \\ 1 & \text{if } x < 0 \end{cases}$$

and $e^0 = 1$, hence e^c stays bounded.

Proof (2/2)

From the hierarchy of infinities theorem we have:

$$\frac{x^{n+1}}{(n+1)!} \rightarrow 0 \text{ as } n \rightarrow +\infty$$

hence the error tends to zero, that is:

$$\frac{x^{n+1}}{(n+1)!} e^c \rightarrow 0 \text{ as } n \rightarrow +\infty$$

since it is the product of an infinitesimal sequence and a bounded one.

We have therefore proved that the function e^x can be written as the sum of a power series, its Taylor series, convergent for every $x \in \mathbb{R}$. □

A new formula for the number e

Observation 2

$$e = \sum_{k=0}^{\infty} \frac{1}{k!} \quad (3)$$

Proof

It suffices to set x equal to 1 in the Taylor series of the exponential function:

$$e = e^1 = \sum_{k=0}^{\infty} \frac{1^k}{k!} = \sum_{k=0}^{\infty} \frac{1}{k!}$$



Two methods to approximate e (1/4)

- ▶ We have derived a second definition of Euler's number e (Napier's constant); the first one was:

$$e = \lim_{k \rightarrow +\infty} \left(1 + \frac{1}{k}\right)^k \quad (4)$$

Two methods to approximate e (2/4)

- Consequently, we have two methods for computing an approximation of the value of e :
 1. The first method is based on formula (4), and by fixing $k = n$ we obtain the approximation:

$$e \approx \left(1 + \frac{1}{n}\right)^n$$

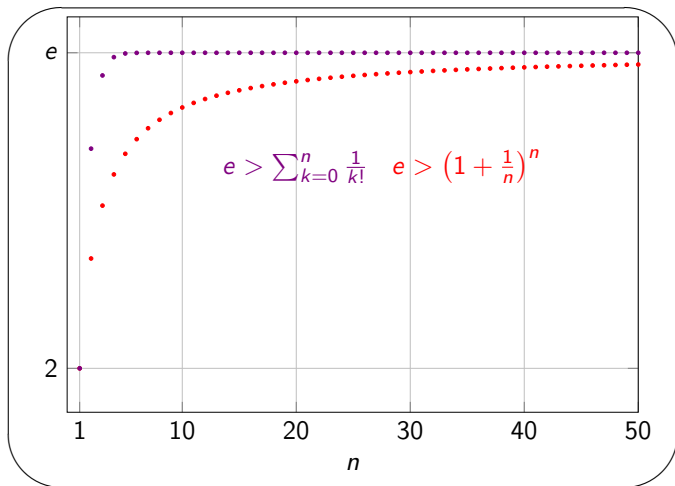
2. The second method is based on formula (3), and by computing the n -th partial sum we obtain the approximation:

$$e \approx \sum_{k=0}^n \frac{1}{k!}$$

Two methods to approximate e (3/4)

	$(1 + \frac{1}{n})^n$	$\sum_{k=0}^n \frac{1}{k!}$	e
$n = 1$	2.0000000000 ...	2.0000000000 ...	2.7182818284 ...
$n = 2$	2.2500000000 ...	2.5000000000 ...	2.7182818284 ...
$n = 3$	2.3703703704 ...	2.6666666666 ...	2.7182818284 ...
$n = 4$	2.4414062500 ...	2.7083333333 ...	2.7182818284 ...
$n = 5$	2.4883200000 ...	2.7166666666 ...	2.7182818284 ...

Two methods to approximate e (4/4)



- The second method is therefore much more efficient at approximating the value of e

Summary

Exponential series:

- ▶ $\forall x \in \mathbb{R}: e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$
- ▶ The remainder $\frac{e^c}{(n+1)!} x^{n+1} \rightarrow 0$: e^c is bounded and $\frac{x^{n+1}}{(n+1)!} \rightarrow 0$
(hierarchy of infinities)

The number e :

- ▶ $e = \sum_{k=0}^{\infty} \frac{1}{k!}$ (setting $x = 1$)
- ▶ The partial sums $\sum_{k=0}^n \frac{1}{k!}$ approximate e much faster than $(1 + \frac{1}{n})^n$

The Taylor series of the elementary trigonometric functions



Introduction: The Taylor series of the elementary trigonometric functions

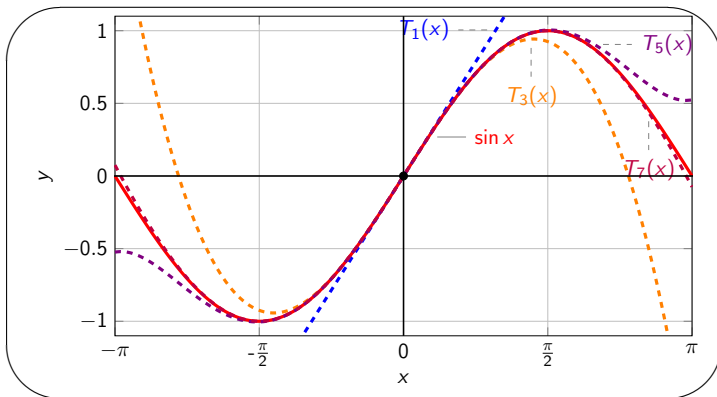
- ▶ In this section we apply the same reasoning to $\sin x$ and $\cos x$
- ▶ We start from the Maclaurin expansions with Lagrange remainder and observe graphically how the Taylor polynomials approximate the two functions better and better
- ▶ We thus obtain the Taylor series of sine and cosine, valid on the whole of $\mathbb{R}$

Example (1/4)

Maclaurin formula/expansion of odd order of the sine with Lagrange remainder:

$$\begin{aligned}\sin x &= T_{2k+1}(x) + \frac{\sin^{(2k+2)}(c)}{(2k+2)!} x^{2k+2} \\ &= x - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots + (-1)^k \frac{x^{2k+1}}{(2k+1)!} \\ &\quad + \frac{\sin^{(2k+2)}(c)}{(2k+2)!} x^{2k+2} \quad \text{with } c \in [0, x]\end{aligned}$$

Example (2/4)

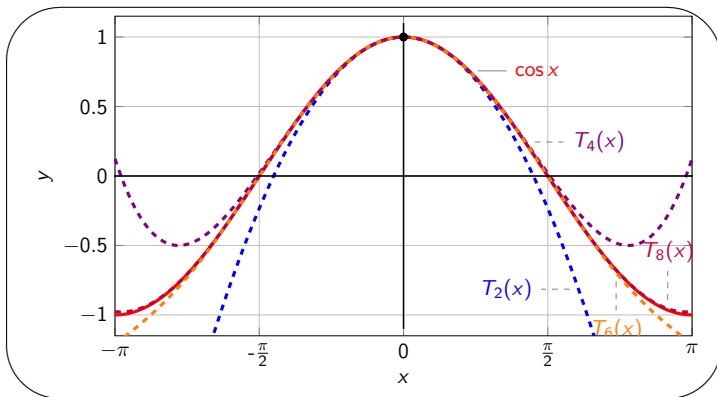


Example (3/4)

Maclaurin formula/expansion of even order of the cosine with Lagrange remainder:

$$\begin{aligned}\cos x &= T_{2k}(x) + \frac{\cos^{(2k+1)}(c)}{(2k+1)!} x^{2k+1} \\ &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots + (-1)^k \frac{x^{2k}}{(2k)!} \\ &\quad + \frac{\cos^{(2k+1)}(c)}{(2k+1)!} x^{2k+1} \quad \text{with } c \in [0, x]\end{aligned}$$

Example (4/4)



Taylor series of sine and cosine

Observation 3: Taylor series of the elementary trigonometric functions

$$\forall x \in \mathbb{R}, \quad \sin x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}$$

$$\forall x \in \mathbb{R}, \quad \cos x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}$$

Proof

The derivatives of $\sin x$ and $\cos x$ are $\pm \sin x$ or $\pm \cos x$, hence for every c :

$$\left| \sin^{(2k+2)}(c) \right| \leq 1, \quad \left| \cos^{(2k+1)}(c) \right| \leq 1$$

For fixed x , the Lagrange errors satisfy:

$$\left| \frac{\sin^{(2k+2)}(c)}{(2k+2)!} x^{2k+2} \right| \leq \frac{|x|^{2k+2}}{(2k+2)!} \rightarrow 0$$
$$\left| \frac{\cos^{(2k+1)}(c)}{(2k+1)!} x^{2k+1} \right| \leq \frac{|x|^{2k+1}}{(2k+1)!} \rightarrow 0$$

as $k \rightarrow +\infty$, by the hierarchy of infinities theorem. Hence the series converge to $\sin x$ and $\cos x$ for every $x \in \mathbb{R}$. □

Summary

Elementary trigonometric series:

- ▶ $\sin x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}$ and $\cos x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}$ for every $x \in \mathbb{R}$
- ▶ The sine contains only odd powers, the cosine only even powers
- ▶ Taylor polynomials of increasing degree approximate the functions on wider and wider intervals

Final summary (1/2)

Taylor series:

- ▶ The Taylor series of f centered at x_0 is $\sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$
- ▶ It converges to $f(x)$ when the Lagrange error $E_n(x) \rightarrow 0$
- ▶ A function may be expandable on the whole of $\mathbb{R}$, on a bounded interval, or only at x_0

Elementary transcendental functions:

- ▶ e^x , $\sin x$, $\cos x$ are expandable in Taylor series on the whole of $\mathbb{R}$

Final summary (2/2)

Formulas to remember:

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}, \quad e = \sum_{k=0}^{\infty} \frac{1}{k!}$$

$$\sin x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}, \quad \cos x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}$$